



Virtual Head-up Displays for Augmented Reality in Cars: a User Testing to Validate the Congruence

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ABSTRACT

Head-up displays (HUD) permit augmented reality (AR) information in cars. Simulation is a convenient way to design and evaluate the benefit of such innovation for the driver. For this purpose, we have developed a virtual HUD that we compare to real AR HUDs from depth perception features. User testing was conducted with 24 participants in a stereoscopic driving simulator. It showed the ability of the virtual HUD to reproduce the perception of the distance between real objects and their augmentation. Three AR overlay designs to highlight the car ahead were compared: the trapezoid shape was perceived as more congruent than the U shape overlay.

CCS Concepts

• **Human-centered computing~Mixed / augmented reality** • *Human-centered computing~User interface design* • *Computing methodologies~Mixed / augmented reality*

Keywords

Automotive head-up display; simulation, virtual environment; depth perception; human machine interface; augmented reality; distance warning.

1. INTRODUCTION

Augmented Reality (AR) is defined by Azuma [1] as any interface that has the following three characteristics: it combines the real objects and the virtual information; it is interactive in real time; and registered in 3D. The latter characteristic means that the virtual information is perfectly superimposed on the real object for the user's eyes.

AR is a promising feature for automotive human-machine interfaces (HMI) as the ocular path to get information is reduced to zero. Foreseeable technical solutions for AR in cars are head-up displays (HUD).

User experiments should be conducted to know if registered AR really increases the efficiency and comfort of the driver. It is easier to conduct such studies in a driving simulator, because simulation enhances repeatability and safety, and allows exploring new use cases (e.g. automated driving [2]) and new technologies beyond their actual limitations.

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Among HUD limitations, there are the size of the field of view, which is limited by the volume available in the dashboard area, and the fact that the image is projected at a fixed distance: whereas the object can be at any distance from the driver's eyes, the virtual information highlighting the objects is always at the same one. One key issue is to design the overlays so that the eyes would nevertheless perceive them as matching their related objects, in other words, the issue is to make the congruence between the objects and their overlays as good as possible. Some HUD prototypes provide a stereoscopic image [3], which allow them to fully respect Azuma's definition, but such systems require a lot of room within the dashboard.

Our need is to be able to study in a driving simulator registered AR made by a HUD that would project at a fixed distance. We first have to validate that it is possible to study the perception of the congruence in the simulation environment, to be able to correctly assume the benefit of AR information for the driver in a real car. We therefore think that congruence should be carefully looked for overlay graphical design perspectives. Some studies already used virtual HUDs [4], [5] to aid HUD design (e.g. projection distance definition), but without fully comparing the virtual HUD with the real ones.

The paper describes first both classical and AR HUDs, and gives some insights about human depth perception as it may explain the differences between real and virtual HUDs. It then reviews the interest of stereoscopic simulation environments and explains how we conceived and developed the virtual HUD for AR. An explicit comparison of real and virtual HUDs is done from depth perception features. The last part describes the user testing that has been conducted to validate the depth perception of the virtual information and the related objects in a simulator, and to compare the congruence of three different overlay graphical designs. Our hypotheses are the following: 1-the virtual HUD is able to reproduce the perception of the depth gap between the object and its overlay; 2-this perception is modulated by overlay graphical design, some shapes being thus more congruent than other ones.

2. REAL HUD

2.1 Classical HUD

A HUD is a device that produces an image, which appears superimposed on a real scene at a given distance for an observer. It is primarily used as an extension of the instrument cluster and gives quick access to formal data like speed, speed limit, GPS orientation and increases the time the driver is focused on the road.

Estimations [7] show that it takes up to 1.1sec for a driver to glance from the road to the cluster and back on the road, and 0.6sec to glance from the road to the virtual image and back to the road. The time the driver eyes are onto the road is thus greatly increased by the use of a HUD [9].

The figure below describes the working principle of a classical HUD for the automotive field:

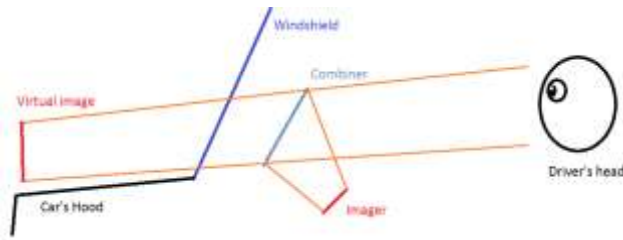


Figure 1. Classical HUD.

An imager (see 0) produces a picture, which is reflected by an optical surface. This optical surface has a complex definition that magnifies the size of the imager and projects its image at a given distance. Currently this distance, called projection distance, is fixed because allowing the device to dynamically modify the distance would drastically increase the price.

For a windshield type HUD, the windshield itself reflects the light coming from the imager, and at least one mirror in the optical system must magnify and compensate the complex shape of the windshield.

At the present time, a TFT (Thin Film Transistor) imager is used in the display¹ and the projection distance is around 2 m; in other words, the virtual image is located at the edge of the hood of the car. Compared to a cluster, it minimizes accommodation effort [9].

The virtual image size is described by the notion of Field of View (FOV). In 2014, most HUDs for cars had a horizontal FOV of around 4.5° and a vertical FOV of around 2°, which will result in a virtual image of 158mm x 70mm at 2m projection distance.

This virtual image is not placed in front of the horizontal gaze of the driver, but at a lower angle, called look down angle. It is more convenient because the virtual image is not overlapping onto the real scene.

2.2 HUD for AR

As far as AR is concerned, some of the set-up of a classical HUD should be reconsidered, especially FOV, resolution and projection distance.

2.2.1 FOV and look-down angle

For AR, the real scene is enriched with virtual data and therefore be ideally at the FOV of the human eye (or at least 20°x10° [3]). But limitations will come from the volume available in the dashboard. For instance, the volume for a windshield HUD with a FOV of 15°x3° is around 3 to 4 times larger than the one with a FOV of 6°x2°.

Concerning the look-down angle, the main interest of an AR system is to superimpose virtual on real environments. Thus, it makes sense to have a look-down angle closer to the horizon. Unfortunately regulation does not allow putting any part above a given line (normalized), which has limited the combiner HUD to a vertical FOV of ~2°. But windshield HUDs do not have this limitation, as they do not put any new parts in front of the driver

because they use the windshield to reflect the image produced by the HUD system.

2.2.2 Resolution

In order to display a sharp image, the resolution of the image shall be close to that of the human eye, which is 1 arcmin. It means that the eye can resolve up to 1 pixel per arcmin, i.e. 60 pixels per degree. But with AR, the visual grammar will probably differ, e.g. using shapes instead of alphanumeric characters. In this case, the resolution of the system could be decreased which helps to simplify the design of an AR HUD system.

2.2.3 Projection distance

Classical HUDs are used as an extension of the cluster and the data do not really need to be aligned to any real object. The information is “floating” in front of the driver and he/she can quickly focus on the virtual image and focus back on the road.

The discussion is different with AR HUD because the virtual image is superimposed on a real object. A projection distance of 2m is of course out of question for an AR system, and it is a complex topic to find the best working projection distance between 8m and 15m.

2.3 Human depth perception

Amongst the several clues for human depth perception, we want to highlight the concepts of focusing and vergence, as long as aerial perspective, head motion parallax [6] and occlusion.

When looking for the sharpest image, the eyes are focusing. The focusing power is measured in diopter (δ , in m^{-1}), which is defined by the reciprocal of the focusing distance in meters. As an example, if the object is seen sharp up to 2 m, the focusing power is 0.5 δ .

Vergence is the ability of each eye to target the same object [6], by modifying the direction of sight. As far as AR systems are concerned, it's an important mechanic because of the possible vergence disparity between the position of the real object and the virtual image that has to highlight this real object.

The aerial perspective is the modification of saturation of color in the long distance. The longer the distance is, the less saturate will appear an object. It's one of the clues used by our vision to estimate position of objects [6].

The head motion parallax effect [6] refers to the relative angular movement of the head with the object. It gives indication to the viewer of its relative position to an object. The longer the distance is, the faster the parallax point of view changes between an object and the observer.

Occlusion (also called ambient occlusion) is a computing technique that creates the illusion of depth by shading the virtual image. It helps the viewer to understand if a virtual image is placed before or after a real object.

All these elements will be discussed further, as their behaviors would vary between a real environment and a simulated one.

3. VIRTUAL HUD

3.1 Objectives

The objectives were to design a virtual HUD module for the SCANeR studio driving simulation software [15] that would allow conducting user experience studies on HUD displays and AR. The requirements that the module should meet were: (i) to display HUD images at the same distance they are perceived by the driver with the real system; (ii) to be able to evaluate different HUD

¹ We are referring to cars like Peugeot 3008, 2014 Mazda 3, 2014 Mini Cooper, 2015 Mercedes class S...

characteristics like distance and FOV; (iii) to be able to display the same kind of images displayed by the real system with similar deviations of colors and transparencies. Also, for the simulation tests to be as close as possible to tests in a real vehicle with the real system, the global system architecture had to be respected.

3.2 Stereoscopic simulation environment

With immersive simulation systems like CAVE or driving simulators with large FOV screens, the use of stereoscopic rendering has already been proven to add valuable realism for the perception of distances [cf. 10, [11]]: in addition to monocular information like shading and perspective, binocular information brought by the computation of distinct images for each eye adds horizontal disparity and thus helps to get closer to the natural vision. The use of dynamic tracking of the head position also adds motion parallax cue which reinforces depth perception.

However, with stereoscopic displays, while the convergence phenomenon, like with natural vision, is targeted on the object, the eye accommodation is focused on the screen and therefore is the same for the whole scene.

3.3 General description

To be able to use the same software and hardware components in virtual and real use cases, the interfaces of the real system were respected: during the virtual tests the object detection is computed with SCANer virtual camera model and delivered to the HMI controller through a dedicated Ethernet protocol and, as with the real system, the computed image is transmitted to the virtual HUD through an HDMI link.

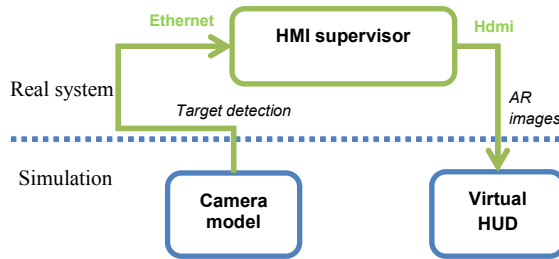


Figure 2. Architecture of the system.

The image is then displayed on an invisible plane in the 3D virtual scene which is computed from the observer point of view and the HUD characteristics: distance, vertical and horizontal FOV sizes. In a stereoscopic environment with the observer's head tracked, the distance of the image is correctly perceived through the binocular disparity and the parallax effect.

Another specific requirement we had to deal with was that the image computed by the HMI controller contains a black background and that transparency gradations are composed of dark color variations. These dark colors appear transparent on the real HUD, because it corresponds to an absence of light emitted by the imager, but would appear opaque on the virtual HUD. To solve this issue the input image is processed in real-time to convert dark colors into transparency information. This processing produces true colors (faithful rendering) and also restores the anti-aliasing of the image content with the background, thus conserving the visual quality. This specific processing, which can be performance consuming, was implemented on the GPU to be able to guarantee an optimal frame-rate.

4. SUMMARY ON THE COMPARISON BETWEEN REAL HUD AND SIMULATED HUD

The following table (Table 1) describes the differences that could be listed between a real HUD and our simulated HUD.

Besides differences to outside conditions, we should also notice that, in the simulator environment, the luminosity of the scene and the annotations are both produced, and therefore limited, by the video projector. The same goes for the resolution. Likewise, the object and the corresponding annotation are both virtual.

5. USER TESTING TO EVALUATE THE VIRTUAL HUD

The purpose of the testing is to check whether the depth differences between overlays, which are projected at a fixed distance by the HUD, and real objects in the driving simulator, are perceptible. We want to be sure that the relative distance between the virtual and the real object is correctly reproduced and perceived in the simulated world. This is to ensure that the studies that will be conducted in the simulation world, such as overlay graphical design, or congruence, usability and utility evaluation, are predictive.

Our hypotheses are the following:

- The virtual HUD is able to reproduce the perception of the depth gap between the object and its overlay.
- This perception is modulated by overlay graphical design, some shapes being thus more congruent than other ones.

5.1 Virtual environment and virtual HUD

The tool used in this experiment to produce the virtual environment was one of Renault's driving simulators, named CARDS (Figure 3). It is a full vehicle mock-up with 240° FOV. It allows virtual stereoscopic vision with Volfonti glasses. The driver is seated at 2.2m from the main screen; the accuracy of focus of each pixel was equal to 0.108°. CARDS has been used for AR HUD studies [5].



Figure 3. CARDS driving simulator.

Table 1. Gap between real and simulated HUD.

Depth clues	Real HUD in real environment	Virtual HUD in stereoscopic virtual environment
As stated in <i>1 – Human Depth Perception</i>	Occlusion, Perspective, head motion parallax effect, shadow, transparency and aerial perspective (colour desaturation at long distance) are the same	
Vergence disparity	8m for annotations (0.13 δ) 20m for vehicle (0.05 δ)	
	Defects brought by the HUD, like image disparity for large FOV (e.g. Both eyes see 2 images that are too different to be merged as only one), flickering, ghost image	No defect, because the vergence is done on the same plane
	Interocular gap for each person, dominant eye is used	[scene and HUD] Interocular gap is fixed by the setup, calculated by the centre of both eyes
Focusing effort	8m for annotations (0.13 δ) 20m for vehicle (0.05 δ) Effort of eyes' accommodation decreased drastically after 6m ([6])	Scene and HUD are seen at 2,2m (0.45 δ). The gap between annotation and real object is greater than 0.25 δ , so the eyes need to accommodate at a near distance.
System and environment brightness	High: - Environment brightness ranges from 10 to more than 10.000 cd/m² - HUD brightness could range from 10 to 15.000 cd/m²	Low: - Brightness would not exceed ~300 cd/m² (assuming projector produces 2000 lm). - The size of the eye pupil is hence small, that increases the depth of field.
HUD brightness	The HUD brightness is added to the scene	The HUD brightness is mixed to the scene (virtual image and simulated scene are produced by the same projector)
Chromatic field	Scene : extended HUD : limited to the gamut of the imager	Scene : limited to the gamut of projector HUD : limited to the gamut of projector
HUD resolution	High (>70 pixels / degree)	Low , and limited by the one of the projector (in our setup, resolution is around ~35 pixels/degree)
Annotated object	Real objects	Virtual objects (the scene and HUD share the same expression)

The virtual HUD had the following dimensions: 15° horizontally, 9° vertically, from 0° to -9°. It is centered along the driver axis. The projection distance is equal to the one of our prototype car, 8m. The driver got the vehicle speed information through a classical HUD simulated by a video projector directed toward the main screen: the distance projection is as such equal to 2.2m, a distance which is similar to the one used in real cars for classical HUDs.

5.2 Simulation Scenarios and Stimuli

A flat motorway with three lanes and a hard shoulder was chosen for a test track. A car in front, later called “target vehicle”, drove in the same lane with a constant speed equal to 15km/h (4.17m/s). This target car was highlighted by an overlay, displayed on the virtual HUD. Through this overlay, the driver was informed about his time headway (THW, which is the distance from the tip to the car to the tip of the target car) by the distance warning system. Three overlay graphical designs were compared: U shape, transparent trapezoid, non-transparent trapezoid later called plain trapezoid. The overlay size is scaled by the HMI supervisor from the target car bounding box dimensions measured by the “target detection” module. The length of the trapezoid is equal to the

distance travelled by the driven car within 1s (4.17m at 15km/h). The transparent trapezoid is similar to the plain one on its edge the closest to the driver; transparency is added on the edge below the tires to reproduce aerial perspective. Our hypothesis is this should increase congruence.

Three distances D were tested: the overlay appeared when the driven vehicle is 5m, 10m or 15m behind the target car. At 5m, THW=1.2s; at 10m, THW=2.4s; at 15m, THW=3.6s. The distance warning system had a tolerance of 2 m, which means that the overlay was displayed when the target car was at D±2m.

The following Figure 4 shows the relative position of target car (here at 5m), overlay (here transparent trapezoid), and of one part of the simulator front screen, when the participant drives exactly at 15km/h at 5m away from the target car.

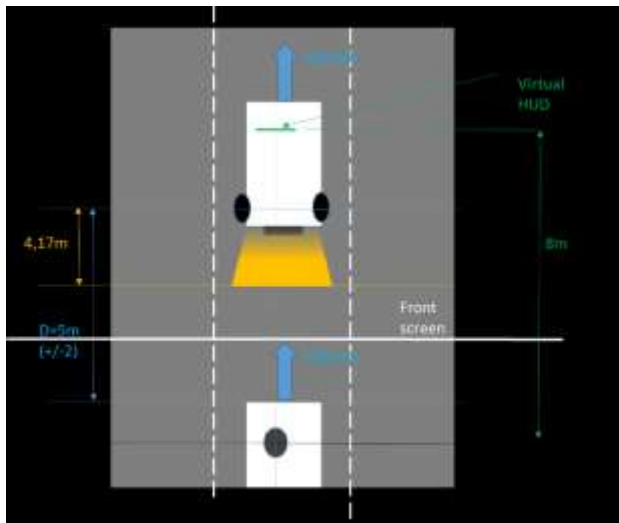


Figure 4. Setup of the experiment in the simulator.

The following pictures (Figure 5, 6, 7) show the target car at 15m with the U shape overlay, at 10m with plain trapezoid and at 5m with the transparent trapezoid (distances $\pm 2m$).



Figure 5. Picture of the simulator showing U shape overlay (target vehicle at 15m).



Figure 6. Picture of the simulator showing plain trapezoid overlay (target vehicle at 10m).



Figure 7. Picture of the simulator showing transparent trapezoid overlay (target vehicle at 5m).

The following distances have been compared: 8m (where the target car is exactly at the same distance as the virtual image), 6m ($8-2$) and 10m ($8+2$), but the differences were too small to be perceived.

A total of $3 \times 3 = 9$ scenes had to be tested by each participant. A Latin-square presentation prevented order effects.

5.3 Participants and protocol

We recruited 24 persons. Their ages ranged from 24 to 55 years ($M=36$, $SD=9$), and their gender was 75% male. They all declared that they could correctly see the stereoscopy of the simulation and the colors.

Each participant had to drive and follow the vehicle at the proper distance to activate the overlay, for an average time of 20 min. He was asked to answer if the overlay was either “in front”, or “on”, or “behind” the highlighted car. The distance reference was the bumper of the target car: the overlay is on the car if there is no gap between the overlay and the bumper. The participant was invited to explain his evaluation. For each stimulus during the simulation, he had the possibility to drive as long as he needed to evaluate the relative distance between the target car and its overlay. As the distance warning in a real car would be displayed, not only in urgent cases, but also when the THW would have correct values (around 2s), it thus reproduced the real world situation, where the driver would not be operating in a time constrain manner.

The following figure (Figure 8) shows the perception of one participant for the target car at 5 m: the U-shape overlay is perceived “in front of the target car, on its tires”.

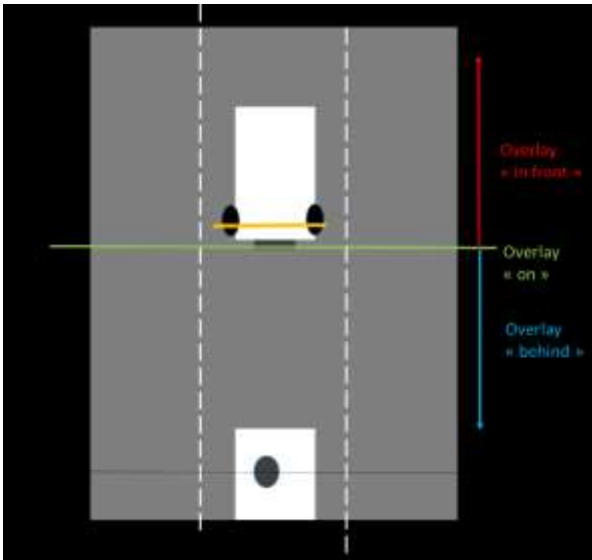


Figure 8. Definition of the position of the overlay: “behind”, “on” and “in front”.

5.4 Test results

The independent variables of the between-subjects design were the distance to the target car and the overlay design. The dependent variable was a categorical one: either in front, or on, or behind the target vehicle.

The following graph shows the number of responses at 5m.

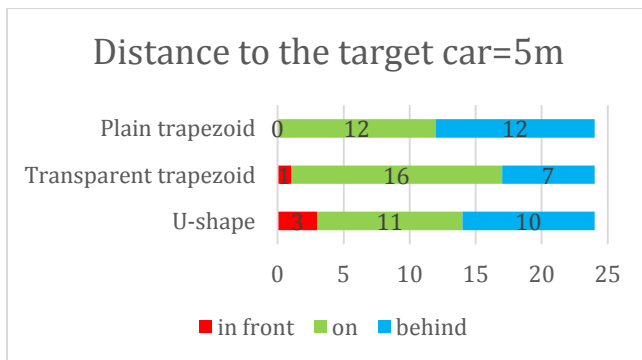


Figure 9. Results for distance to target car is 5 m.

The Fisher’s exact test, which relevant level of significance is determined by $\alpha=0.05$, shows that there is no influence of the overlay design (2-sided $p=0.24$). Nevertheless it indicates that the plain trapezoid is significantly less seen in front of the target vehicle.

Generally speaking, the overlay is mainly seen at the same distance as the target car. In other words, the congruence between the object and its overlay is good at 5m. We were concerned that the overlay would be perceived as “in front”, because it is projected at 8m, but few people had this perception. The participant who perceived the transparent trapezoid in front of the car said that the edge was 10cm below the car; the three participants who perceived the U shape in front of the car said that it was on the tires. As such, the relative distance was small.

At 10m, the results are the following:

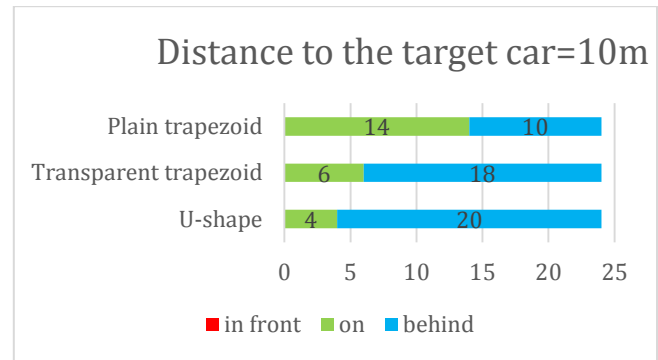


Figure 10. Results for distance to the target car is 10 m.

The Fisher’s exact test shows that there is a significant influence of the overlay design ($p=0.007$). It also indicates that the U shape overlay is significantly less congruent and the plain trapezoid is more congruent than the other ones. The U shape and the transparent trapezoid are rather seen behind the target car.

At 15m, the results are the following:

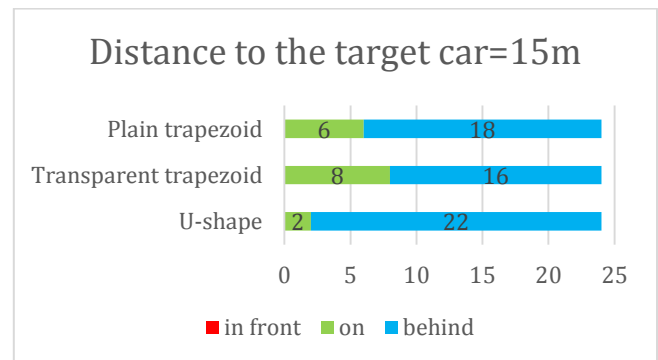


Figure 11. Results for distance to the target car is 15 m.

The Fisher’s exact test shows that the influence of the overlay design is low ($p=0.099$). It shows the U shape overlay is significantly less congruent.

The number of persons who perceived the transparent trapezoid behind is similar to the one at 10m. At 15m, the three overlay designs are mostly perceived behind the car. Some participants were able to quantify the relative distance between the target car, the overlay and their eyes. These verbatim quotes indicate that the U shape overlay is perceived as “far away from the target car” by 11 persons (2 for the transparent and 4 for the plain trapezoid), whereas 2 perceived it as “close”.

The measure of the resistance of the overlay design toward the distance was done using Fisher’s exact tests. The U shape is strongly influenced by the distance ($p=0.0008$); the transparent trapezoid is influenced ($p=0.05$) and the plain trapezoid is slightly influenced ($p=0.0554$); the test also indicates that this overlay is more perceived “behind” at 15m than at 5 or 10m.

Verbatim analysis provided information which visual cues participants used to evaluate the relative distance, for the different simulations. Occlusion is most cited (11 participants), second (5) is shadow (if the participant could see a shadow between the target car and the overlay, he/she concluded the overlay was behind).

Some drivers explained which overlay they would prefer for a distance warning system: 5 preferred the transparent trapezoid over the plain one, 2 the transparent one over the U shape. Three

complained that the plain trapezoid hid the registration plate of the target car.

5.5 Discussion

Our first hypothesis (“the virtual HUD is able to reproduce the perception of the depth gap between the object and its overlay”) is validated because all participants were able to answer to the question “is the overlay in front / on / behind the target car?”. The additional and optional question “could you quantify the distance” asked to help the participants explaining their evaluation “the overlay is behind”, is more difficult: 7 persons said at least once throughout the test that they could not tell.

Our second hypothesis (“the perception of the depth gap is modulated by overlay graphical design, some shapes being thus more congruent than other ones”) is validated. The U shape overlay is the least congruent as it is mostly perceived “behind” from 10m (20 participants), with an increasing number of participants at 15m (22, 11 from them evaluating the distance as large). The plain trapezoid is the most congruent as it starts to be perceived as behind only at 15m. The transparent one is less congruent than the plain one at 10m. One can give the following explanation: the transparency on the edge that touches the car creates a gap and allows a dark shadow behind the car to be seen. The transparent trapezoid holds up well at 15m, so that it is similar to the plain one at that distance.

On the road, the distance to the real objects is generally higher than 15m: e.g., at a speed of 90km/h (25m/s) the target car should be at 50m (considering the common 2s delay rule of thumb for safety distance). As such, it would be interesting to compare the transparent and plain trapezoids at higher distances and check if our hypothesis “transparency helps congruency” will be confirmed.

The trapezoid overlay seems promising to facilitate a distance warning system because it provides a better congruence than the U shape one. Some drivers of our test said that they preferred the transparent trapezoid. One should conduct user testing to study the usefulness and usability of the trapezoid overlay for a distance warning system. To check usability, the questions could be: is it distracting? Is the trapezoid annoying because of its surface? Should only critical levels be displayed? Sun [13] explored the distribution of visual attention and gave some evidence of competition between tasks where stimuli are not spatially separated, which is the case for the overlay and the environment. The distracting effect of AR for the driver has been confirmed by Charissis [4] and thus should be checked. Schwerdtfeger [14], on maintenance tool application, saw that overlay graphical design should be carefully done to avoid occlusion of real object. Since 3 participants of our test complained about this, it should also be tested. To check usefulness the question could be: does AR allow the driver to maintain larger THW, or does it incite to take more risks (like increase the vehicle speed [8])?

For AR-HUD design, the following can be concluded from our tests: Even if the overlay is projected in front of the car, it is never seen in front of the tires. The projection distance of 8m can thus be chosen even if, in very dense traffic conditions, the real object may be closer. One major cue to trick the eye in this situation is the occlusion. Our objective will be to find out which graphical cues will eliminate the relative distance perception, in other words tricking the driver’s eyes, when the real object is far behind the overlay.

6. CONCLUSIONS AND PERSPECTIVES

Throughout the paper, an explicit comparison of real and virtual HUDs is done from depth perception features.

Our experiment shows that the virtual environment is able to reproduce relative distance perception between the objects and their overlays. As such, user testing can be conducted in the virtual stereoscopic environment with the virtual HUD we developed to benchmark AR designs, particularly to find out which ones ensure good congruence to the real objects. The trapezoid shapes seem to be more congruent than the U shape: next experiments on real cars will be conducted to check that this evaluation is predictive.

Congruence is not the only feature that meets AR: usability and usefulness should be carefully studied. Simulation tools are very suitable to evaluate and compare different overlay designs, especially for new driving systems like automated driving, to see how AR can help the driver to have good awareness of what the system detects and, while taking over, to properly restore his awareness of the environment.

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